

CS 310 : Automata Theory

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Minimization of DFAs

Let A be a DFA with no unreachable states

Let $\approx \subseteq Q \times Q$ be the state equivalence relation (computed say by the table-filling algorithm)

$$p \approx q \Leftrightarrow \forall x \in \Sigma^* (\hat{\delta}(p, x) \in F \Leftrightarrow \hat{\delta}(q, x) \in F)$$

- ▶ $\approx$ is an equivalence relation with finitely many classes
- ▶ Let $[q] = \{q' \mid q' \approx q\}$

Minimization of DFAs

Given DFA $A = (Q, \Sigma, \delta, q_0, F)$, we can minimize A to the DFA $A_{min} = (Q', \Sigma, \delta', q'_0, F')$ called the *Quotient Automata* where

- ▶ $Q' = \{[q] \mid q \in Q\}$
- ▶ $\delta'([q], a) = [\delta(q, a)]$ for all $a \in \Sigma$
If $p \approx q$ then $\delta(p, a) \approx \delta(q, a)$. $[p] = [q] \Rightarrow [\delta(p, a)] = [\delta(q, a)]$
- ▶ $q'_0 = [q_0]$
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$q \in F$ iff $[q] \in F'$. If $p \approx q$ and $q \in F$, then $p \in F$. Each class is either inside F or disjoint from F .

Minimality of A_{min}

Irreducibility Lemma

A_{min} is irreducible : that is, no two states of A_{min} are equivalent. In other words, if $p \not\approx q$ in A (i.e. $[p] \neq [q]$), then $[p]$ and $[q]$ are *not* equivalent as states of A_{min} .

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- ▶ Recall also $[q] \in F' \Leftrightarrow q \in F$
- ▶ Combining these: $\hat{\delta}'([p], x) \in F' \Leftrightarrow \hat{\delta}(p, x) \in F$, and likewise for q .
- ▶ Since exactly one of $\hat{\delta}(p, x), \hat{\delta}(q, x)$ is in F , exactly one of $\hat{\delta}'([p], x), \hat{\delta}'([q], x)$ is in F' .

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- ▶ So $[p] \not\approx [q]$ in A_{min} . **Distinct states of A_{min} are always distinguishable.**

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- ▶ Since B has fewer states than A_{min} , at least two *distinct* states p, q of A_{min} are equivalent to the *same* state of B .
- ▶ Then $p \approx q$ **contradicting the Irreducibility Lemma**

Uniqueness of A_{min}

Minimality says A_{min} has the *fewest* states. What if some other DFA B matches that count exactly?

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- ▶ This gives a map $f : Q_{A_{min}} \rightarrow Q_B$, sending each $[q]$ to an equivalent state of B .

Uniqueness of A_{min} : the isomorphism

- ▶ f is injective: if $f(p) = f(q)$ for $p \neq q$, transitivity of $\approx$ gives $p \approx q$ contradicting the Irreducibility Lemma applied to A_{min} .

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- ▶ Since $|Q_{A_{min}}| = |Q_B|$, an injective map between equal size finite sets is a bijection.
- ▶ The B -state equivalent to each $[q]$ is also *unique*: else two states r_1, r_2 in B would satisfy $r_1 \approx [q] \approx r_2$, hence $r_1 \approx r_2$ contradicting the Irreducibility Lemma applied to B .

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- ▶ f preserves
 - ▶ the start state ($f(q'_0) = q_{0,B}$, both being the states reached by ε),
 - ▶ preserves acceptance ($[q] \in F' \Leftrightarrow f([q]) \in F_B$, since equivalent states agree on $x = \varepsilon$), and
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 - ▶ commutes with transitions ($f(\delta'([q], a)) = \delta_B(f([q]), a)$, since equivalence is preserved one letter at a time).
- ▶ f is an isomorphism: any DFA achieving the minimum state count is structurally identical to A_{min} , up to renaming states. The minimal DFA is unique.

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- ▶ But this starts from an already-built DFA A for L via Jaffe, or by hand, or however else
- ▶ **Moving on:** can we build the minimal DFA directly from L itself ?

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- ▶ $\sim_L \subseteq \Sigma^* \times \Sigma^*$ is an equivalence relation on words
- ▶ Consider the equivalence classes of $\sim_L$
- ▶ When can $\sim_L$ have only finitely many equivalence classes?

Properties of $\sim_L$

- ▶ $\sim_L$ is a **right congruence** : that is, for any $a \in \Sigma$, $x \sim_L y \Rightarrow xa \sim_L ya$
- ▶ $\sim_L$ **refines** L : that is, $x \sim_L y \Rightarrow x \in L \Leftrightarrow y \in L$

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Myhill-Nerode relations

An equivalence relation on Σ^* is called a *Myhill-Nerode* relation for $L \subseteq \Sigma^*$ if it is a right congruence refining L , and has finitely many equivalence classes. The relation was proposed by John Myhill and Anil Nerode.

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- ▶ So $a^m \not\sim_L a^n$ whenever $m \neq n$.
- ▶ Hence $[a^0], [a^1], [a^2], \dots$ are all pairwise distinct classes.
- ▶ Infinitely many equivalence classes.

Another Example : Consider $L = a^*b^*$

For any word $u \in \Sigma^*$, its **residual** is $u^{-1}L := \{w \in \Sigma^* \mid uw \in L\}$, the set of continuations that land in L .

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- ▶ x does not match a^*b^* : somewhere in x , **an a occurs somewhere after b** .

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 - (ii) or $j \geq 1$ and $x \in a^*b^+$.
- ▶ x does not match a^*b^* : somewhere in x , **an a occurs somewhere after b** .
- ▶ Take the *first* such violation

Another Example : Consider $L = a^*b^*$

For any word $u \in \Sigma^*$, its **residual** is $u^{-1}L := \{w \in \Sigma^* \mid uw \in L\}$, the set of continuations that land in L .

$x \sim_L y$ iff x, y have the same residual: $x^{-1}L = y^{-1}L$.

Every word falls into exactly one of three shapes

Any $x \in \Sigma^*$ either matches the pattern a^*b^* , or it does not.

- ▶ x matches a^*b^* : $x = a^i b^j$ for some $i, j \geq 0$.
 - (i) Either $j = 0$ and $x \in a^*$
 - (ii) or $j \geq 1$ and $x \in a^*b^+$.
- ▶ x does not match a^*b^* : somewhere in x , **an a occurs somewhere after b** .
- ▶ Take the *first* such violation
 - (iii) everything before it is $a^i b^j$ ($i \geq 0, j \geq 1$) and everything after the violating a is arbitrary, so $x \in a^*b^+a\Sigma^*$.

Every word in Σ^* falls into exactly one of these three shapes.

Another Example (continued)

Compute the residual for a representative of each shape:

(i) $x = a^i$: $xw \in L \Leftrightarrow w \in a^*b^*$.

- ▶ Same residual a^*b^* for every $i \geq 0$
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(ii) $x = a^i b^j$ ($j \geq 1$): $xw \in L \Leftrightarrow w \in b^*$ (w cannot have a 's).

▶ Same residual b^* regardless of $i \geq 0, j \geq 1$

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(iii) $x = a^i b^j a z$: xw is *never* in L , for any w

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The three residuals a^*b^* , b^* , $\emptyset$ are pairwise different, giving 3 distinct classes. As every word falls into one of the three shapes, $\sim_L$ has exactly 3 classes.

Myhill-Nerode Relations

If L is regular, then there exists a Myhill-Nerode relation $\sim_A$ for L defined from the DFA A for L .

- Let $A = (Q, \Sigma, \delta, q_0, F)$ be the DFA with no inaccessible states such that $L = L(A)$.

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If L is regular, then there exists a Myhill-Nerode relation $\sim_A$ for L defined from the DFA A for L .

- ▶ Let $A = (Q, \Sigma, \delta, q_0, F)$ be the DFA with no inaccessible states such that $L = L(A)$.
- ▶ Define $x \sim_A y$ iff $\hat{\delta}(q_0, x) = \hat{\delta}(q_0, y)$.
- ▶ It can be seen that $\sim_A$ is a right congruence refining L .
- ▶ $[x] = \{y \mid \hat{\delta}(q_0, x) = \hat{\delta}(q_0, y)\}$. There is one equivalence class corresponding to each state in Q . Hence, finitely many classes.
- ▶ Hence, $\sim_A$ is Myhill-Nerode.

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- ▶ We know $\sim$ is a right congruence refining L , with finitely many equivalence classes $[x]$.
- ▶ From $\sim$, define a DFA $A_{\sim} = (Q, \Sigma, \delta, s, F)$ where
$$Q = \{[x] \mid x \in \Sigma^*\}$$
$$s = [\epsilon]$$
$$F = \{[x] \mid x \in L\}$$
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 - ▶ " $[x] \in F$ " does not depend on which representative witnesses it
- ▶ $x \in L$ iff $[x] \in F$ (definition of F , using the previous point).
- ▶ $L(A_{\sim}) = L$: $x \in L(A_{\sim})$ iff $\hat{\delta}([\epsilon], x) \in F$ iff $[x] \in F$ iff $x \in L$.